Introduction to the Programme

Networks and Innovation Training Programme

Dr Antoine Vernet & Dr Anne ter Wal

September 11, 2013

About us

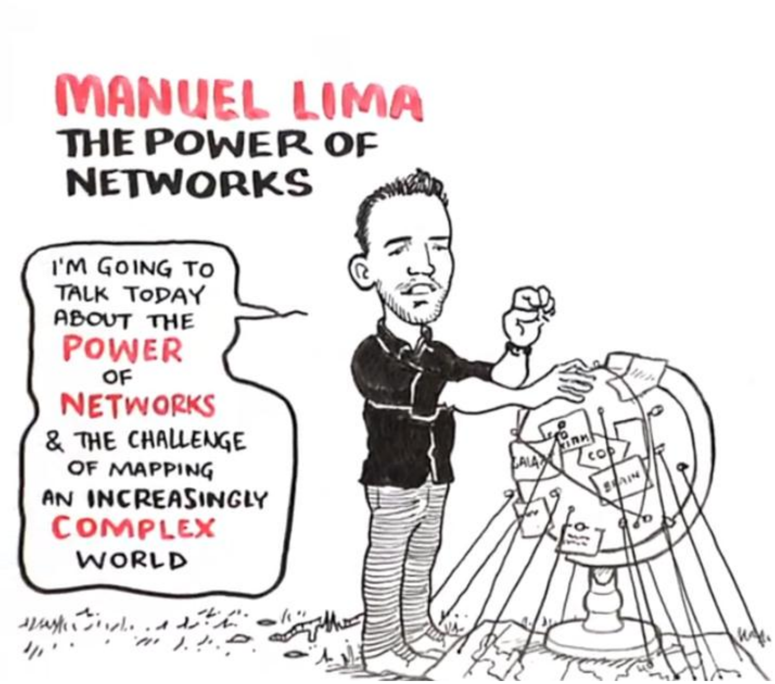
Anne ter Wal

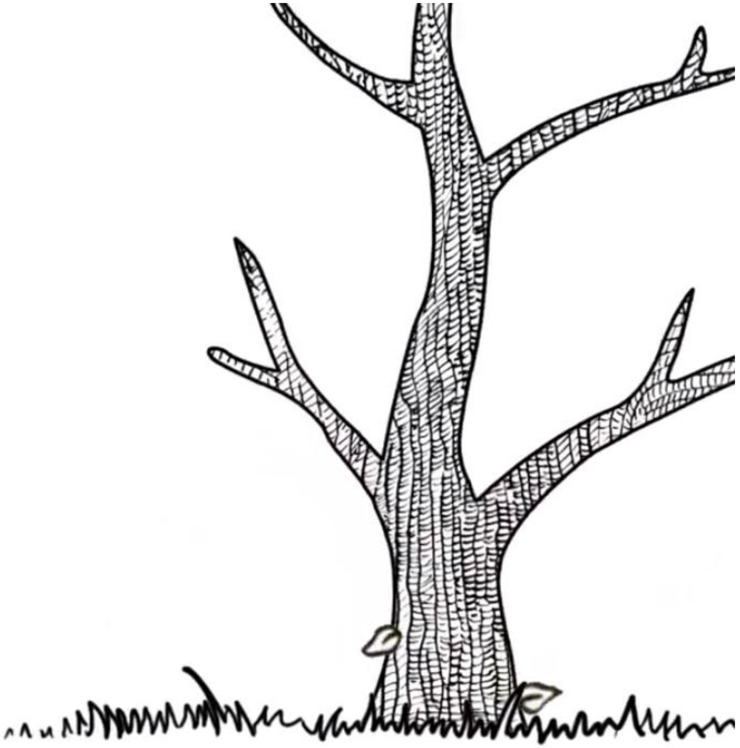
Assistant Professor

Antoine Vernet

Research Associate

The network perspective of looking at the world





Networks as the 'locus' of innovation

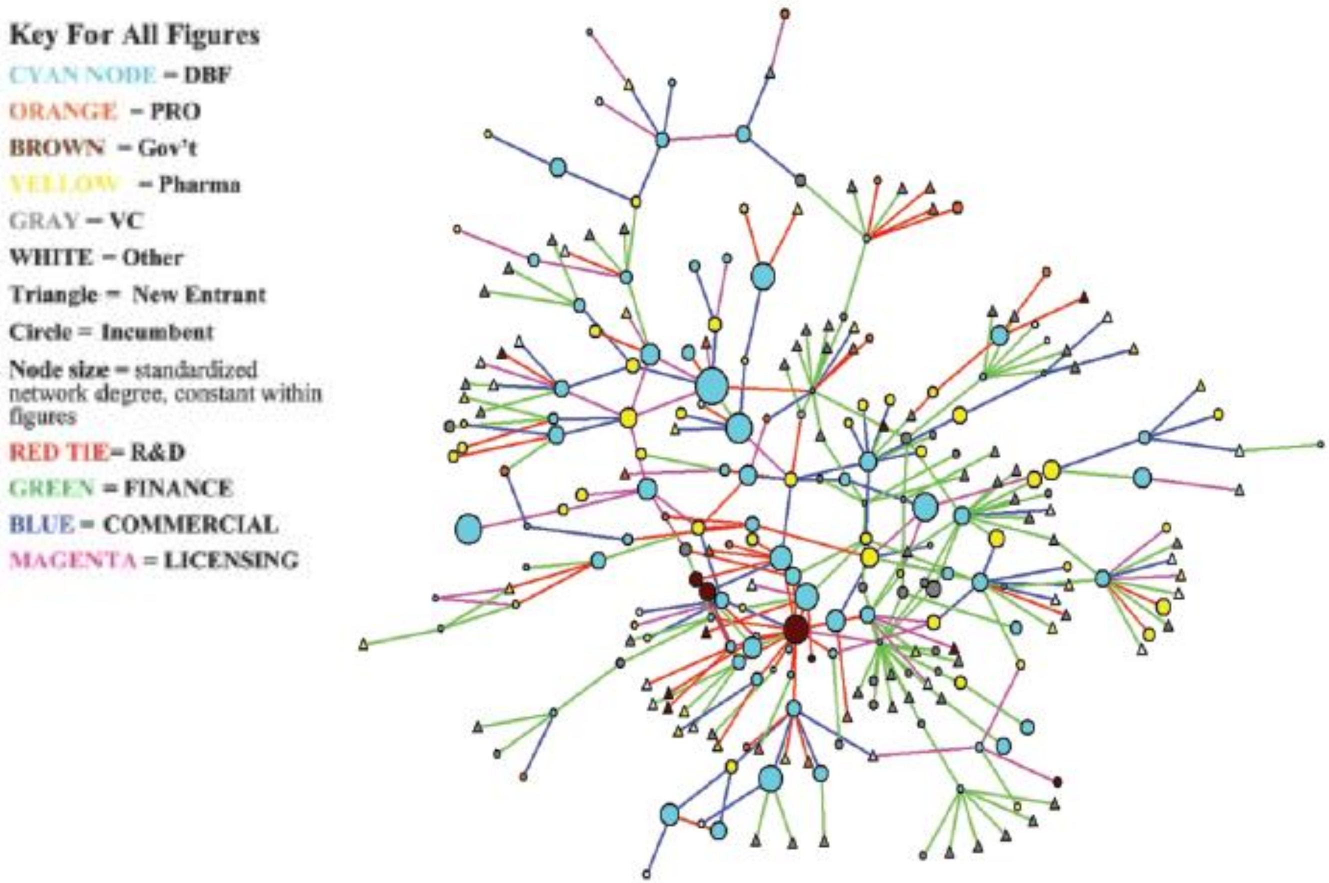


FIG. 5.—1989 main component, new ties

Source: Powell et al (1996, ASQ), Powell et al. (2005, AJS)

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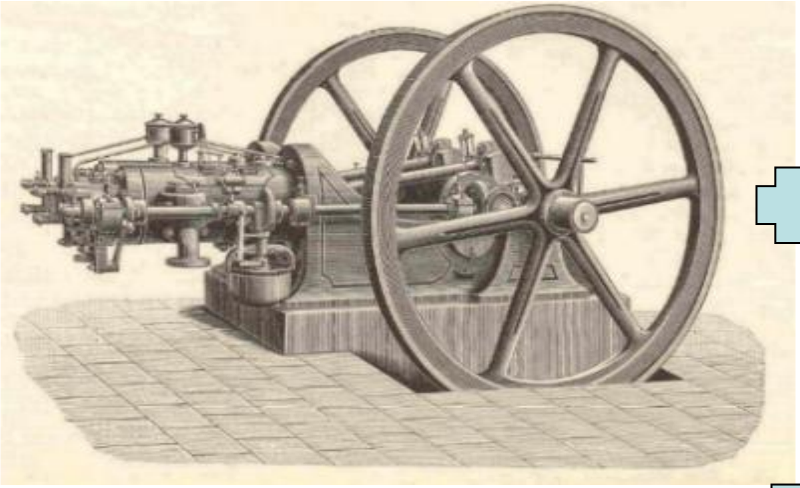
"Sources of innovation do not reside exclusively inside firms; instead, they are commonly found ***in the interstices*** *between firms, universities, research laboratories, suppliers, and customers"*

Powell et al. (1996), p. 118

"A firm does not innovate in isolation, but depends on ***extensive interaction*** *with its environment"*

Fagerberg et al. (2004), p. 20

Innovation as ‘recombination’



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Innovation as ‘recombination’

Innovation as ‘recombination’

(12) **United States Patent**
Armstrong

(54) **USER-OPERATED AMUSEMENT APPARATUS FOR KICKING THE USER’S BUTTOCKS**

(76) Inventor: **Joe W. Armstrong**, 306 Kingston St., Lenoir, TN (US) 37771-2408

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

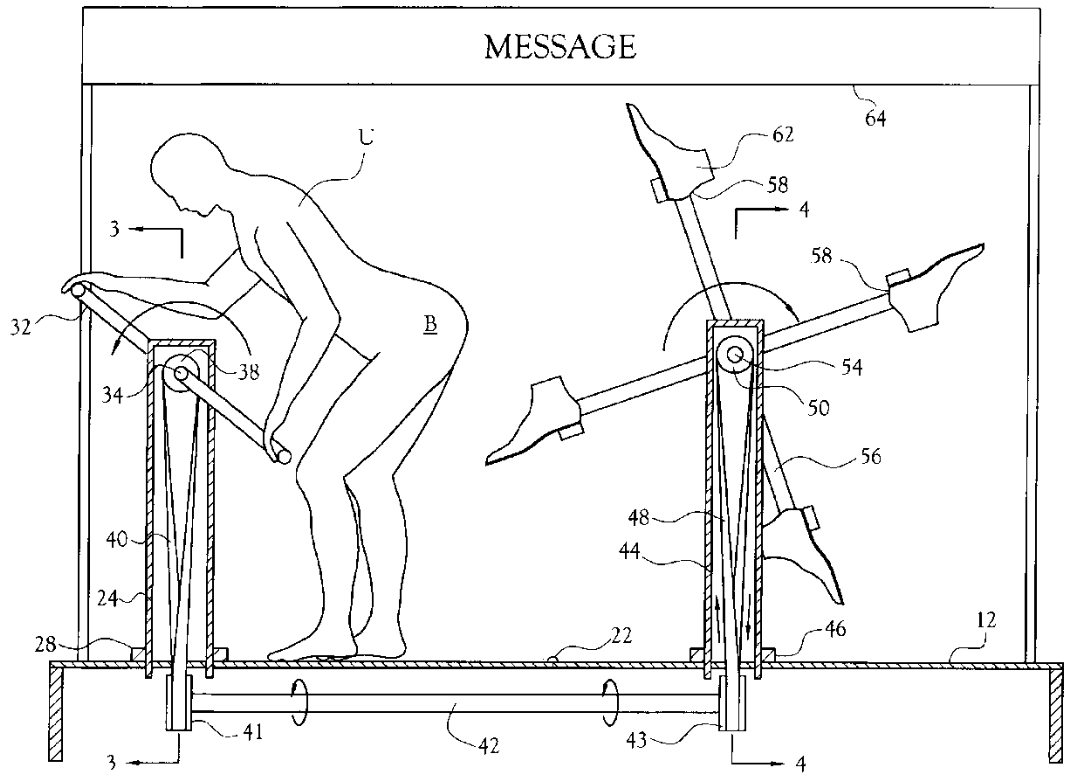
(21) Appl. No.: **09/477,175**

(22) Filed: **Jan. 4, 2000**

(51) **Int. Cl.**[7] **A63H 37/00**

(52) **U.S. Cl.** **472/51**; 472/55

(58) **Field of Search** 472/51, 55, 137; 482/51, 72, 148



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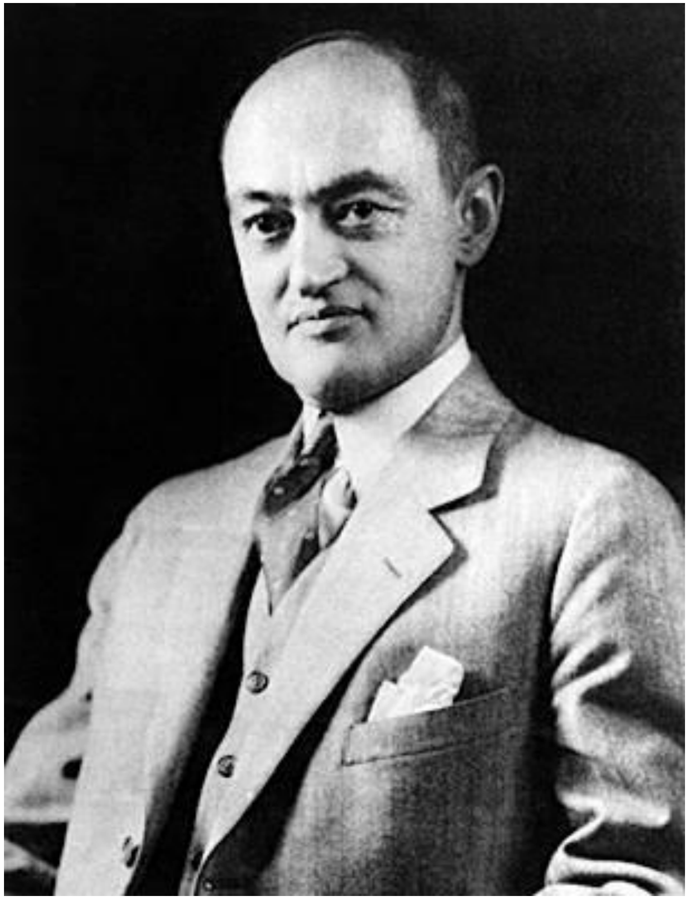
A network approach to innovation

- **Joseph Schumpeter:** innovation is the outcome of **recombination of prior knowledge and ideas**.
- Innovations are increasingly an outcome of **collaborative action** or team work.
- **Networks** consisting of collaborating people that differ in their knowledge, skills and ideas form the **seedbed** in which the recombination process takes place.



Further reading: Becker, Knudsen & March (2006, ICC), Fleming & Sorenson (2004, SMJ), Powell, Koput & Smith-Doerr (1996, ASQ), Katila & Ahuja (2002, AMJ)

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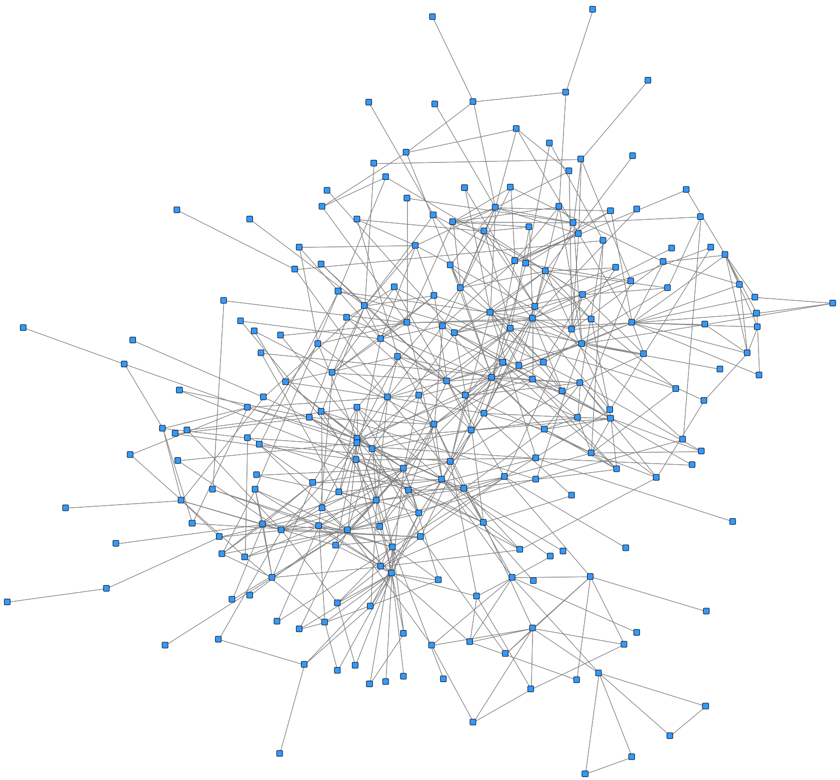
Networks within organizations

A Network Approach to Innovation

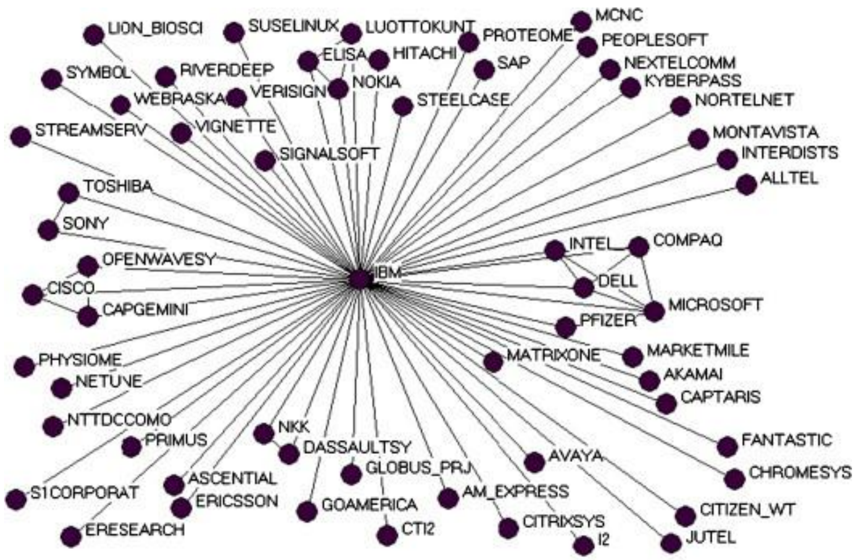


- Knowledge sharing and advice seeking among individuals
  - Borgatti & Cross (2003, Man Sci)
  - Reagans & McEvily (2003, ASQ)
  - Cross & Sproull (2004, Org Sci)
- Collaboration and knowledge sharing in teams / projects
  - Soda, et al. (2004, AMJ)
  - Hansen et al. (2005, AMJ)
- Knowledge transfer across organizational units
  - Hansen (1999, ASQ)
  - Tsai (2001, AMJ)

Source: own research – advice-seeking network among 119 engineering consultants Criscuolo et al. (2013, WP).

Networks between organizations

A NETWORK APPROACH TO INNOVATION

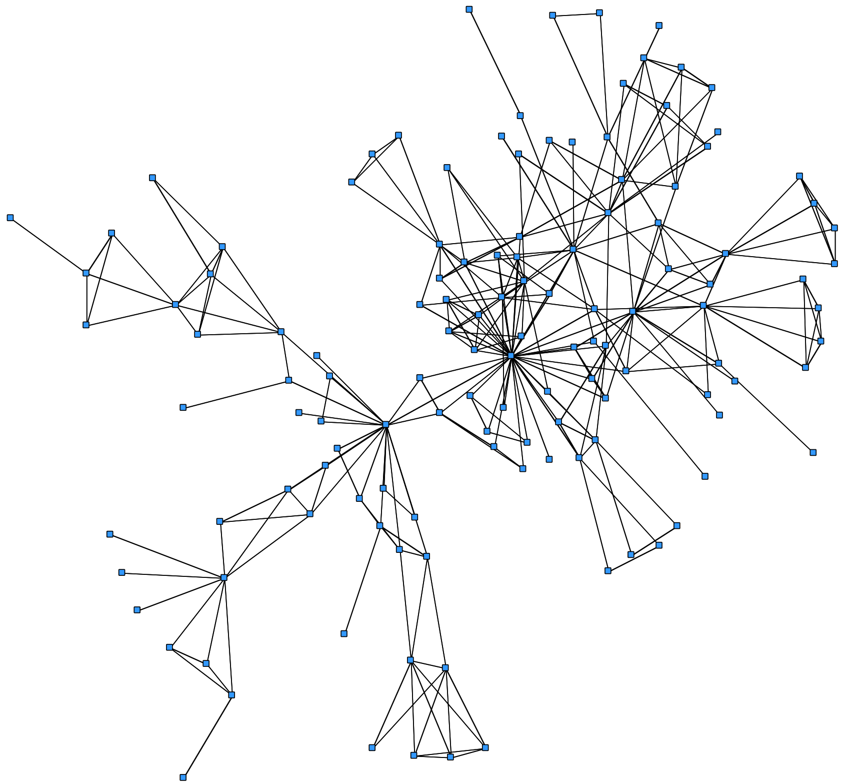


IBM Strategic Alliance Network 2001-2002

- Inter-firm strategic alliances
  - Gulati (1995, ASQ)
  - Ahuja (2000, ASQ)
  - Baum et al. (2000, SMJ)
  - Verspagen & Duysters (2004, Technovation)
  - Schilling & Phelps (2007, Man Sci)
- Other forms of inter-firm collaboration
  - Powell et al. (1996, ASQ)
  - Rowley et al. (2000, SMJ)
  - Shipilov & Li (2008, ASQ)

Networks among individuals

A NETWORK APPROACH TO INNOVATION



Main component of co-inventor network in Life Sciences, Sophia-Antipolis

- Inventor networks
  - Fleming et al. (2007, ASQ)
  - Ejermo & Karlsson (2007, Res Pol)
  - Paruchuri (2010, Org Sci)
  - Lee (2010, Org Sci)
- Networks in creative sectors
  - Uzzi & Spiro (2005, AJS)
  - Perry-Smith (2006, AMJ)
  - Cattani & Ferriani (2008, Org Sci)
  - Zaheer & Soda (2009, ASQ)
- Communities
  - Guimerà et al. (2005, Science)
  - Fleming & Waguespack (2007, Org Sci)

Source: own research (Ter Wal 2013, Reg Studies)

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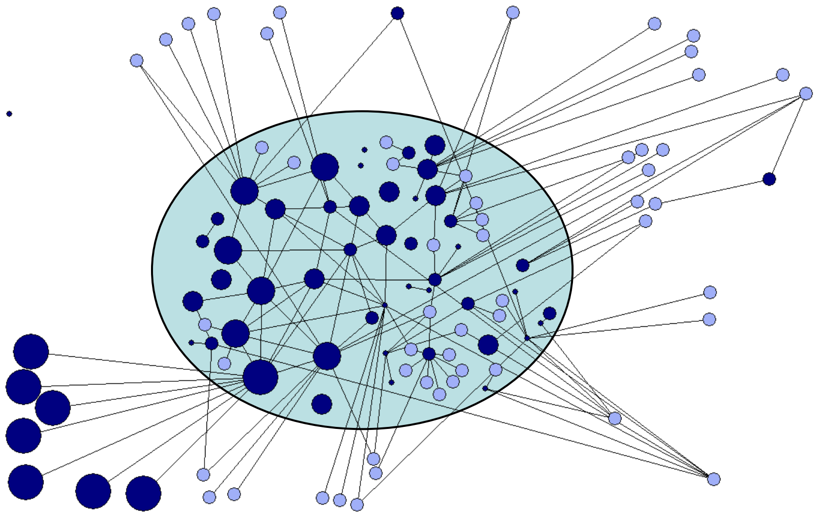
Networks and geography

A NETWORK APPROACH TO INNOVATION



Inter-firm collaboration network of firms in the industrial district of shoe production in Barletta, Italy

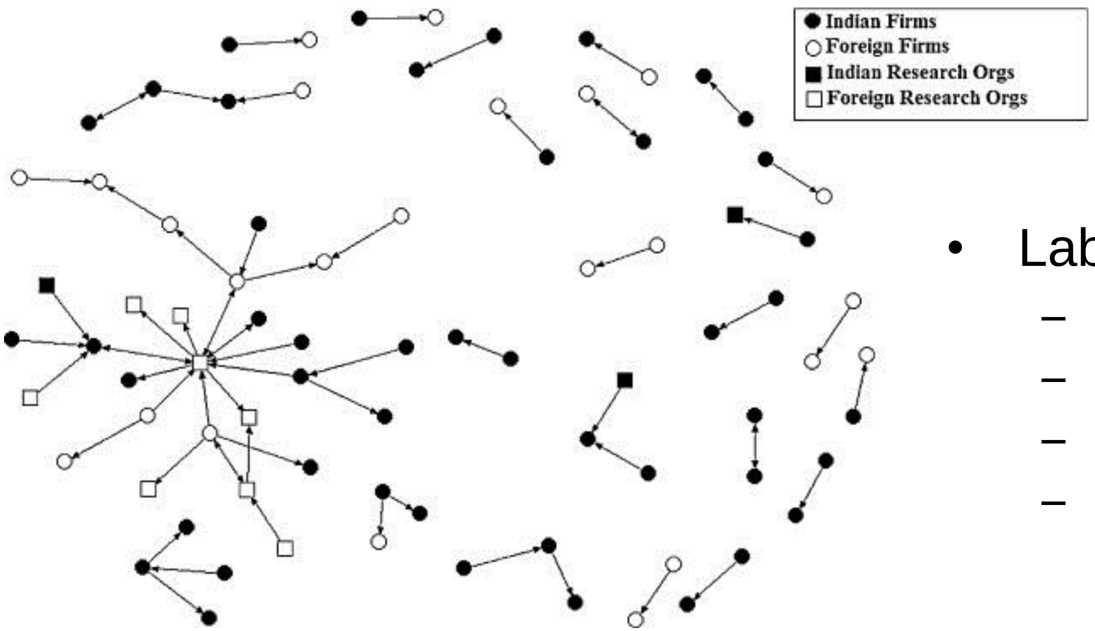
- Networks within and across clusters
  - Giuliani & Bell (2005, Res Pol)
  - Ter Wal & Boschma (2007, I&I)
  - Morrison (2008, Reg Stud)
  - Whittington et al. (2009, ASQ)

- The role of networks in localized knowledge spillovers
  - Owen-Smith & Powell (2004, Org Sci)
  - Singh (2005, Man Sci)
  - Breschi & Lissoni (2009, JEG)

Source: own research (Boschma & Ter Wal 2007, Industry & Innovation)

Labour mobility networks

A Network Approach to Innovation



Mobility network among a set of Indian firms, 1984-2004

- Labour mobility networks
  - Podolny & Baron (1997, ASQ)
  - Agrawal et al. (2006, JEG)
  - Song et al. (2003, Man Sci)
  - Singh (2005, Man Sci)

Source: Alnuaimi, Opsahl & George (2012, Res Pol)

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Programme objectives

At the end of this course, we hope you will be able…

- …to critically assess **existing theories** that explain the role of **networks** in the generation of **innovations**
- …to identify **novel directions for research** at the interface of networks and innovation
- …to apply the **toolkit** of network theories, concepts and methods to **design effective networks research** in your own field of interest
- …to **perform social network analysis** in **R** to examine processes of tie formation, network dynamics and their impact on innovation.

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How networks drive innovation

Networks and Innovation Training Programme
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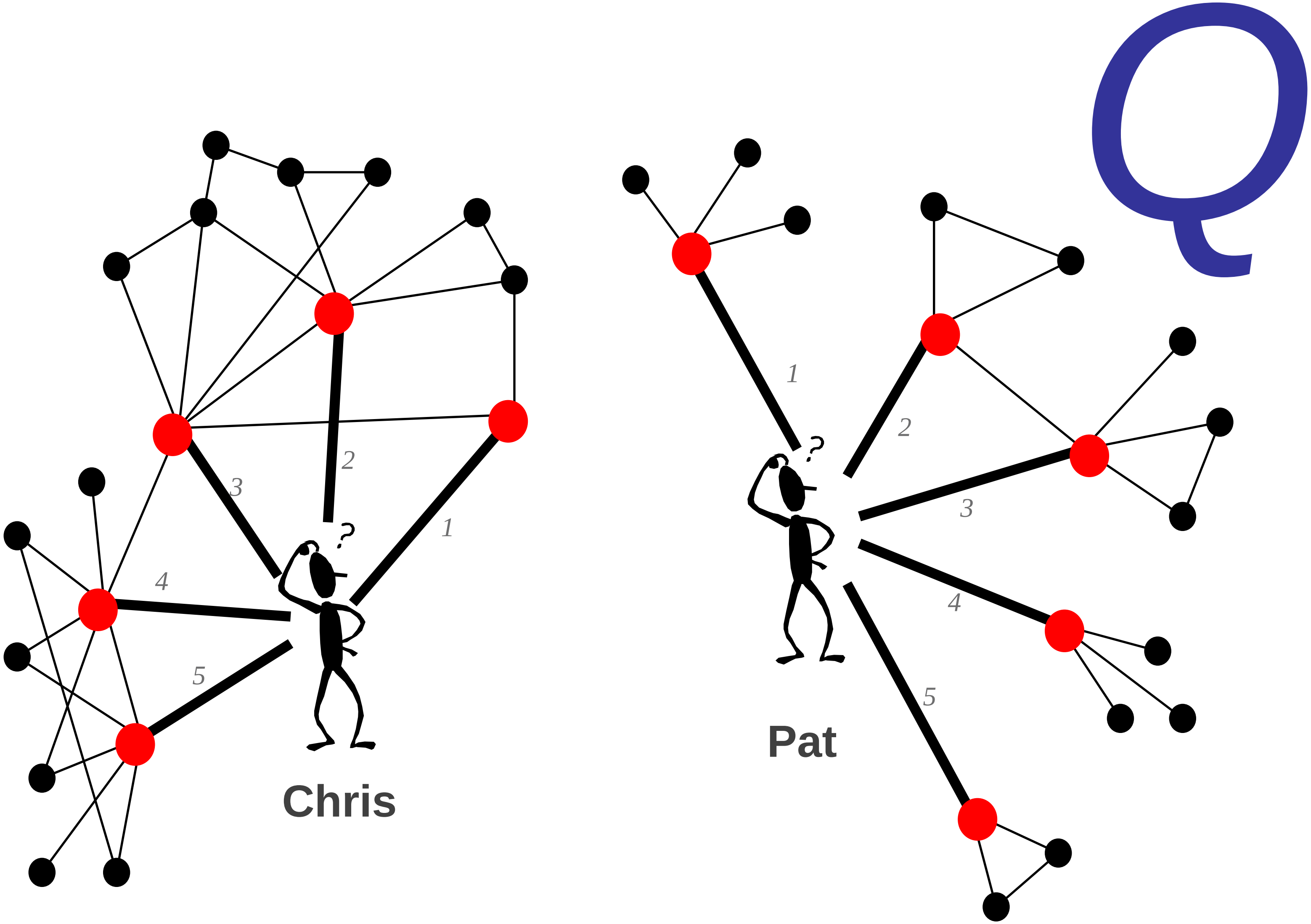
Who is in a better position to innovate?



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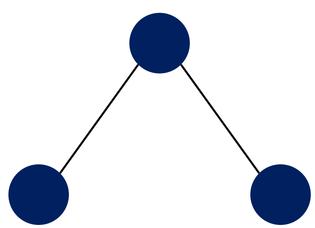
Q

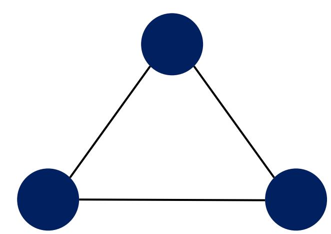
Who is in a better position to innovate?

A



Brokerage

B

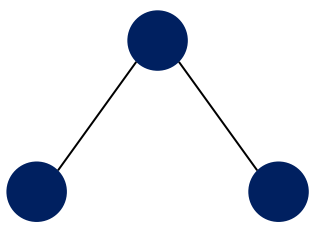


Closure

Q

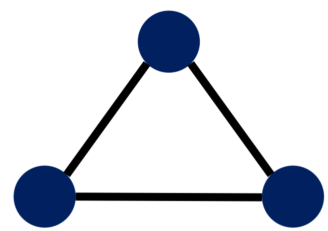
Who is in a better position to innovate?

A



Weak ties

B



Strong ties

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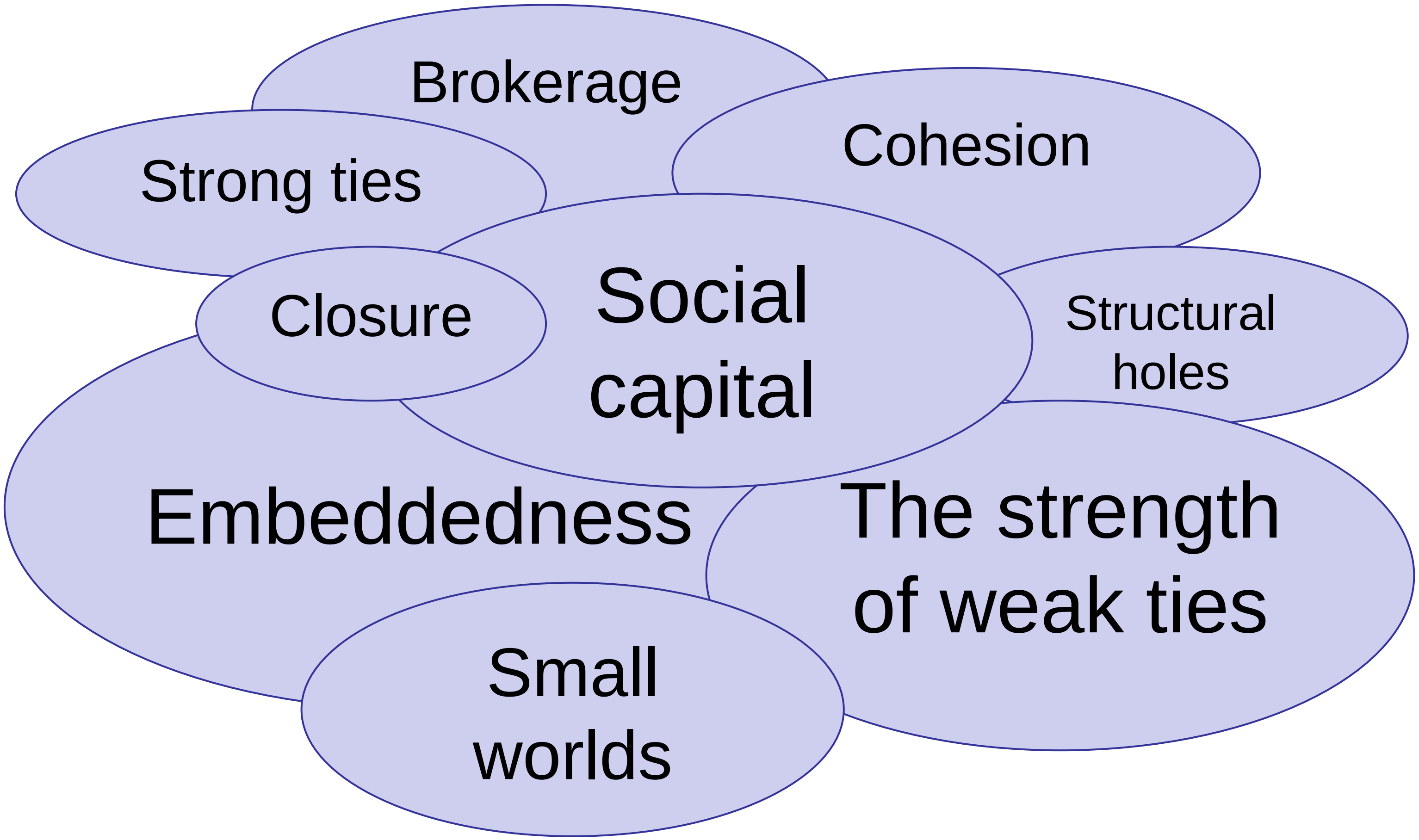
Learning objectives

At the end of this lecture, we hope you will be able…

- …distinguish a range of existing concepts and arguments that explain how **networks** may **impact** on **innovation**
- …position these in the **major theoretical perspectives** and **research traditions** of the field of networks research
- …describe examples of the **current state-of-the art in the field**
- … with the ultimate aim of **positioning your research** against existing theories and state-of-the-art in the field

Concepts


Brokerage
Cohesion
Strong ties
Closure
Social capital
Structural holes
Embeddedness
The strength of weak ties
Small worlds

The baseline

The advantages of **direct ties** for **innovation output**:

- **Knowledge sharing:** each partner can potentially receive a greater amount of knowledge from a collaborative project than it would obtain from a comparable research investment made independently.
- **Complementarity:** collaboration facilitates bringing together complementary skills from different firms.
- **Scale economies:** increasing returns mean that outcomes are greater if partners pool their resources.

Source: Ahuja (2000, ASQ)

The baseline

The advantages of **indirect ties** for **innovation output**:

- **Access to knowledge spillovers:** Inter-firm linkages can be a channel of communication to access the knowledge of partners' partners.

- **Complementarity:** Indirect ties therefore increase a firm's catchment area for information:
  - Information-gathering device, e.g. promising technological trajectories
  - Information-processing device, e.g. a filter through which to absorb, sift and classify information.

THE MAGNITUDE OF BENEFITS FROM INDIRECT TIES ARE RELATIVELY LOW.

Source: Ahuja (2000, ASQ)

Strength of weak ties

THEORETICAL PERSPECTIVES

The stronger the tie between two people, the more likely their social worlds will overlap.

Bridging ties are a potential source of novel ideas, because a person can hear things not already circulating among friends.

This can explain why people hear about new jobs through acquaintances rather than friends.

Source: Granovetter (1973), Borgatti & Halgin (2011, Org Sci)

Embeddedness

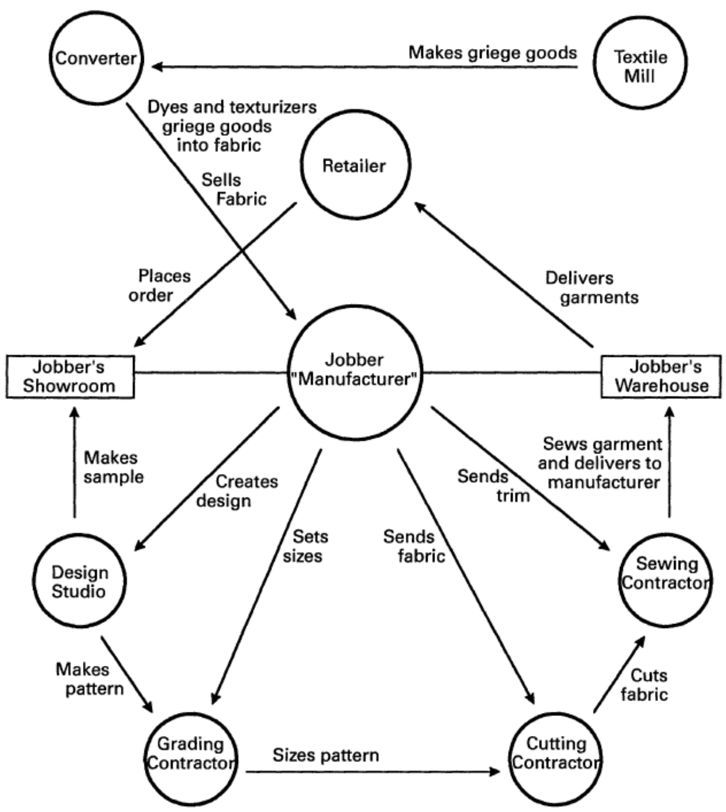
Theoretical Perspectives

Economic action is **embedded** in a network of interpersonal relations.

Firms embedded in networks of exchange relationships have **higher chances of survival** than firms that only maintain arm's length (market) relationships.

Embeddedness is a **logic of exchange** that promotes complex adaptation.

Figure 1. Typical interfirm network in the apparel industry's better-dress sector.



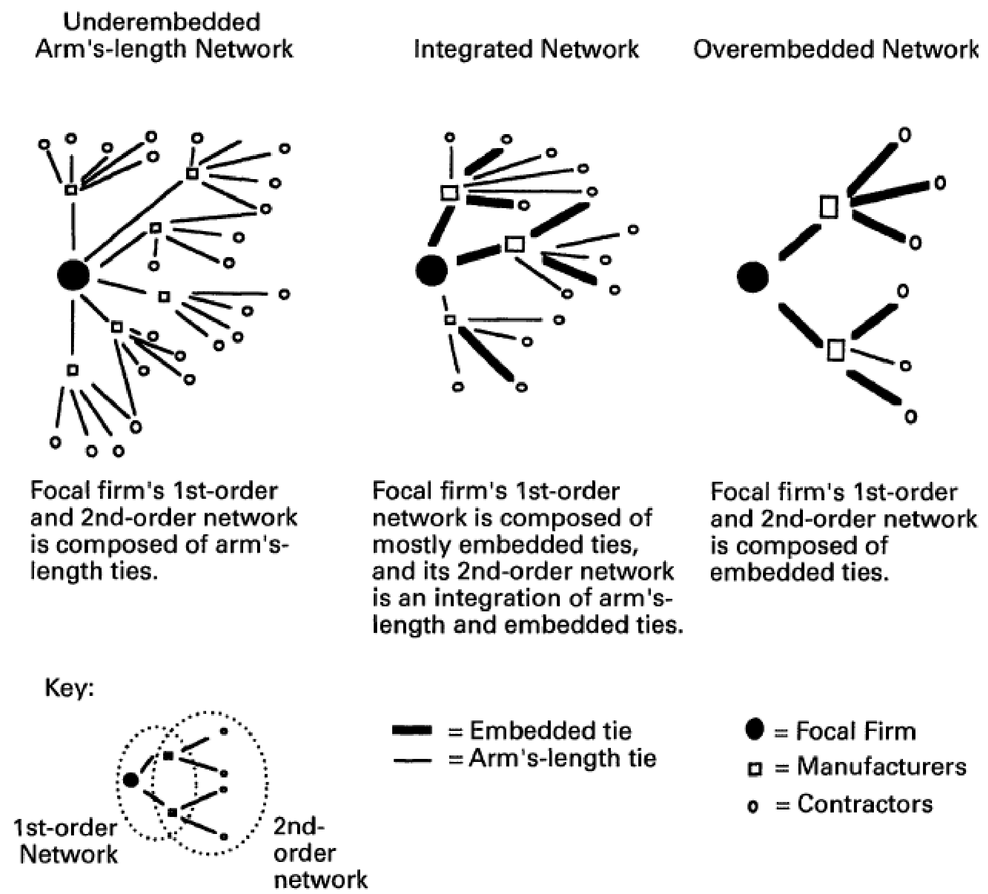
Source: Granovetter (1985), Uzzi (1996/1997)

Embeddedness

THEORETICAL PERSPECTIVES

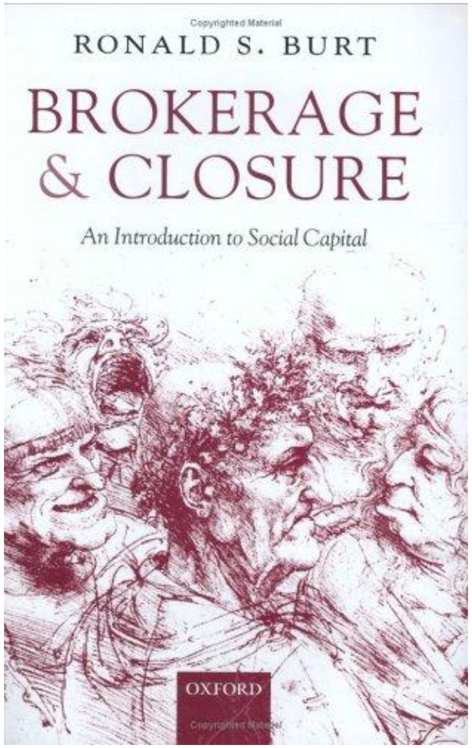
Overembeddedness makes firms…

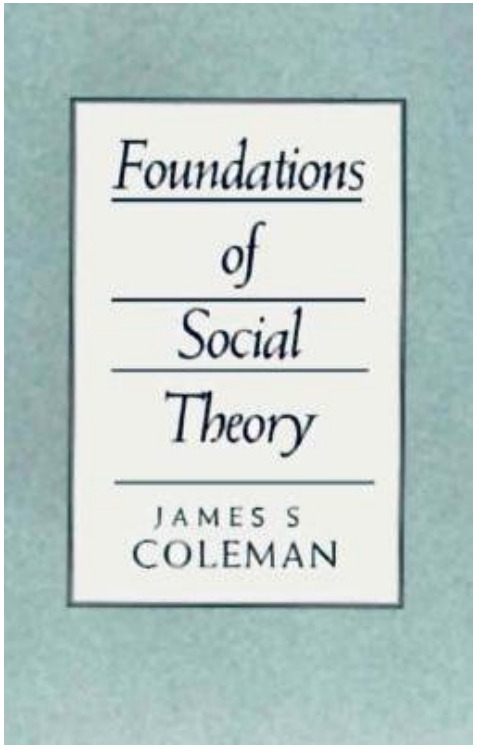
- …vulnerable to exogenous shocks
- …and insulates them from information that exists outside their network.



Source: Granovetter (1985), Uzzi (1996/1997)

Social capital
THEORETICAL PERSPECTIVES





Weak ties?

Strong ties?

Source: Burt (2005), Coleman (1990)

Brokerage

Theoretical Perspectives

Opinion and behaviour are more homogeneous within than between groups.

Brokerage across structural holes provides a vision of options otherwise unseen.

New ideas emerge from selection and synthesis across the structural holes between groups.

Closure

Theoretical Perspectives

If all firms in an industry held relationships with each other, inter-firm information flows would lead quickly to established norms of collaboration.

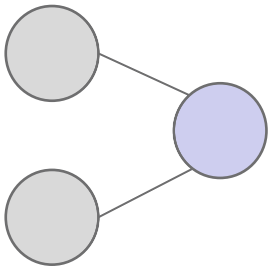
In such a dense network, information on deviant behaviour would be readily disseminated and the behaviour sanctioned.

As the predictability of behaviour is increased, self-seeking opportunism is constrained and the formation of trust enabled.

Source: Walker et al. (1997, Org Sci), see also Coleman (1990), Ahuja (2000, ASQ)

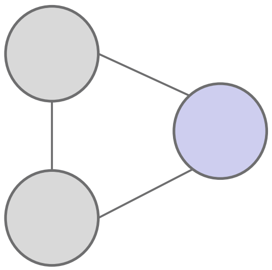
Social capital

THEORETICAL PERSPECTIVES



Brokerage

- Informational advantages
  + Access to unique knowledge
  - Difficult to verify quality
- Control advantages
  - "Tertius gaudens" – use information to own benefit
    + Low cost of maintenance
    - Brokerage positions temporary



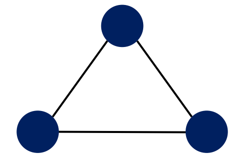
Closure:

- Informational advantages
  + Potential for triage
  - Redundant information / lock-in
- Embeddedness advantages
  - Flow of knowledge to mutual benefit
    + Increased commitment of alters
    - Commitment is costly

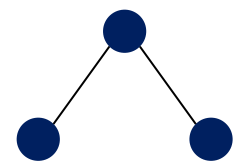
Social capital

THEORETICAL PERSPECTIVES

- Closure as 'social capital'



  - Core advantages:
    - trust-based, intensive knowledge exchange
    - possibility to cross-check information through indirect channels



- Structural holes as 'social capital'
  - Core advantages:
    - efficient knowledge transmission throughout the network
    - unique access to knowledge for nodes at bridges
    - the power to act at the "nexus" of diverse information

Source: Reagans and Zuckerman (2008, ICC)

So…

Q

Are weak-strong tie theory

&

brokerage-closure theory

just the same thing??

Not quite… Yet, they share their core principle!

Strong/weak ties vs. brokerage/closure

Theoretical Perspectives

The value of **non-redundancy** is key to the strength of weak ties and brokerage.

The value of **redundancy** is key to benefits of embeddedness/strong ties and closure/cohesion.

Granovetter/Uzzi	**Burt**
First and second-order networks	Ego networks
Tie strength	Alter-alter ties
Serendipitous ties that incidentally prove useful	Strategic ties that are built with a purpose in mind

Source: Kilduff & Brass (2010), Borgatti & Halgin (2011)

But…

…that doesn’t quite solve the problem.

Are strong ties or closed structures conducive to innovation?

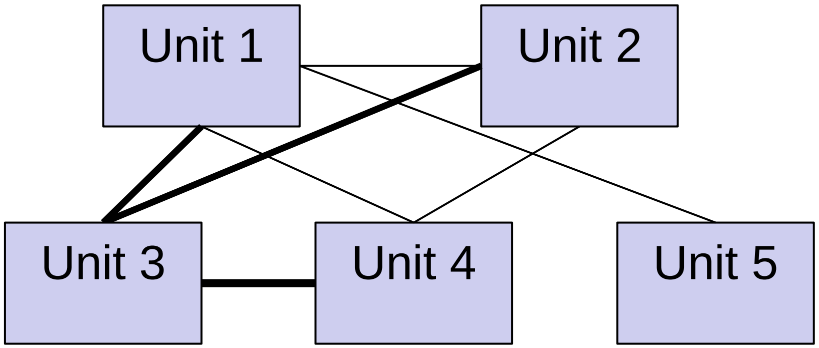
Or rather weak ties or brokered structures?

Let’s have a look at the evidence.

Strength of weak ties: example

Theoretical Perspectives

Which unit would be best placed to generate innovations based on knowledge held in other units?



Weak ties speed up project when knowledge is not complex.

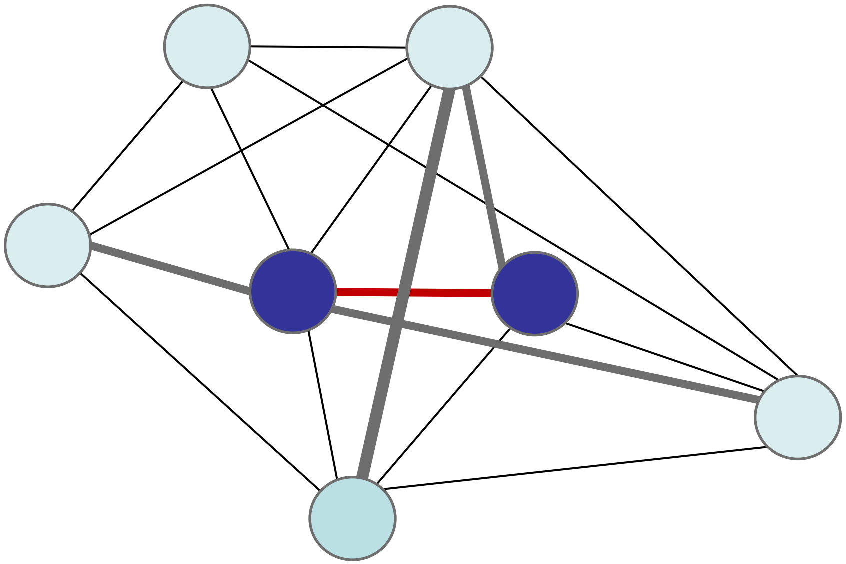
Strong ties speed up projects when knowledge is complex.

Source: Hansen (1999, ASQ) – Example for illustrative purposes only

Embeddedness: example

THEORETICAL PERSPECTIVES

Social cohesion around a relationship affects the motivation and willingness of individuals to invest time, energy, and effort in sharing knowledge with others.



Eases knowledge transfer over and above the effect of tie strength.

Source: Reagans & McEvily (2003), Image for illustrative purposes

Brokerage: example

THEORETICAL PERSPECTIVES

Ethnography of product design firm IDEO

Working for clients in 40 different industries

Brokering ideas between applications aids innovation

Individual consultants who occupy brokerage positions among their clients are in the best position to introduce existing technological solutions in new industries.

Source: Hargadon & Sutton (1997, ASQ)

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Brokerage vs. closure: example

THEORETICAL PERSPECTIVES

Inter-firm collaboration in the international chemicals industry

Bridging structural holes negatively impacts innovation performance

Source: Ahuja (2000, ASQ)

Brokerage vs. closure: example

Theoretical Perspectives

Collaborative brokerage can aid the **generation** of an idea but hamper its **diffusion** and use by others.

First-order negative effect of cohesion on creativity

But marginal benefits of cohesion for creativity if there is inflow of diverse perspective through other means

Breadth of experience

Worked in multiple organizations

Collaborators with external connections

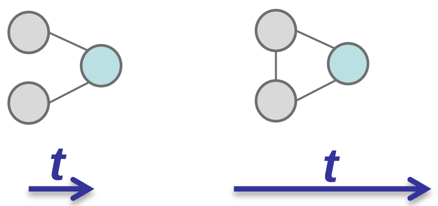
Source: Fleming et al. (2007, ASQ)

Brokerage versus closure

Recent Advances in the Debate

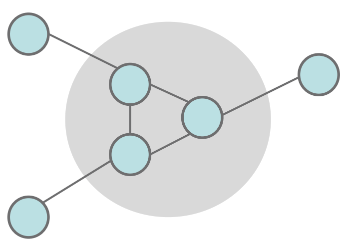
Advantages of brokerage and closure may differ in timing.

(Soda et al. 2004; Zaheer & Soda 2009; Baum et al. 2012)



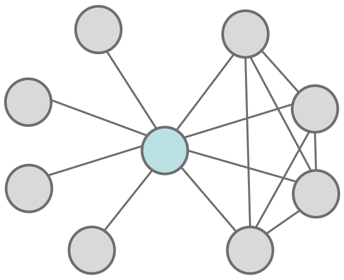
Advantages of brokerage and closure may operate at different levels

(Zaheer and Bell 2005)



Brokered and closed structures can co-exist concurrently and at the same level.

(Reagans and McEvily 2008)

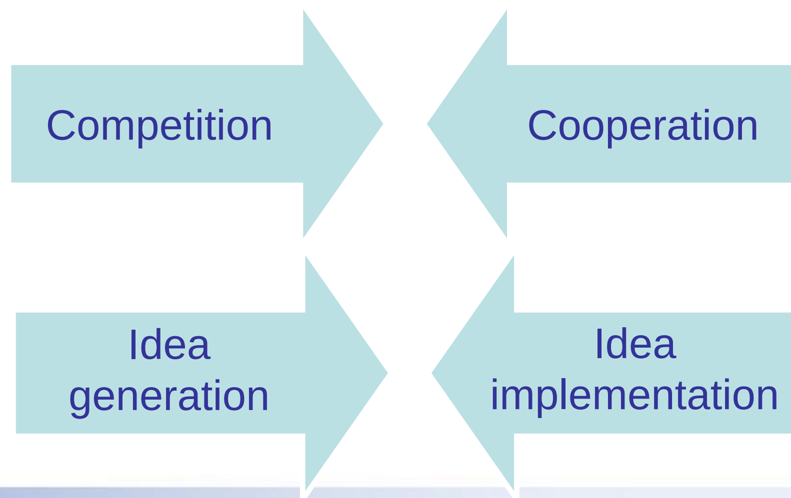


Brokerage vs. closure: summary

Theoretical Perspectives

“Social network analysis should be rooted in the **specifics of time and place**”.

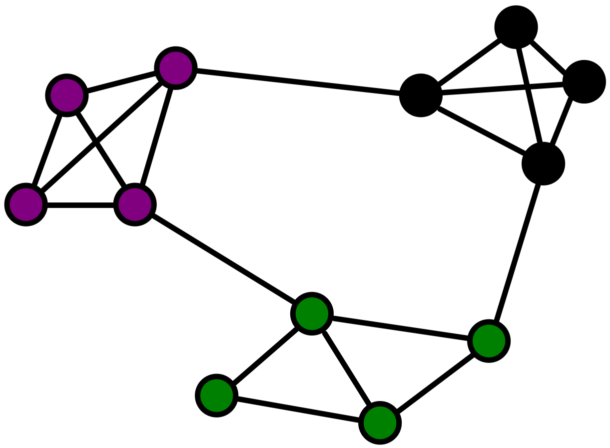
“The question is no longer which perspective is right or wrong, but which measure is appropriate given **the particular context being studied**”

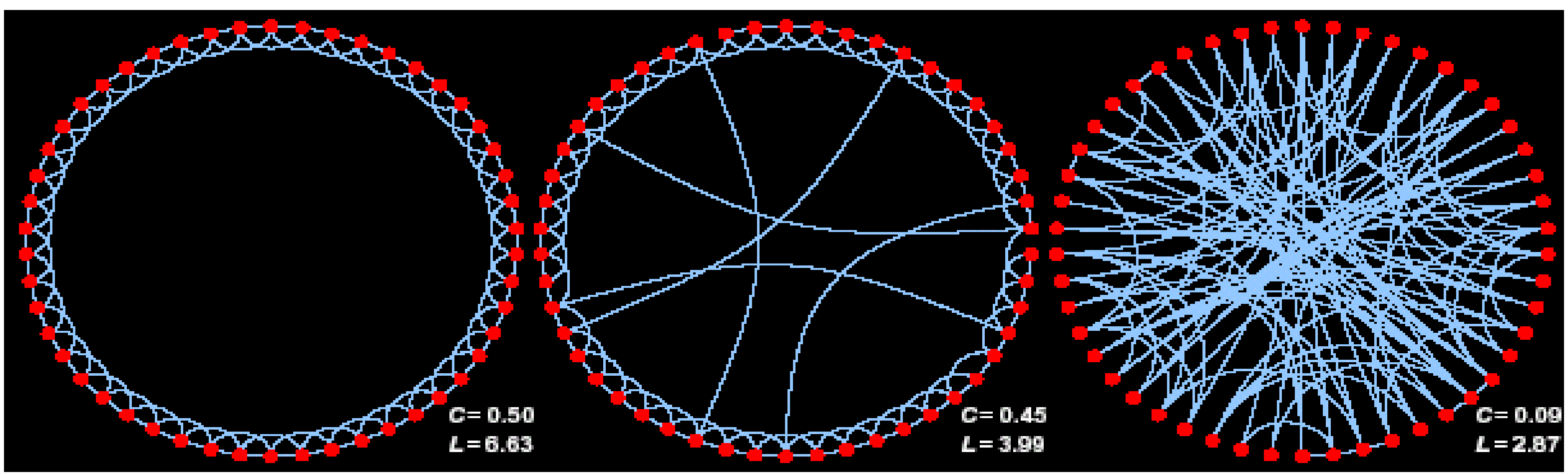


Source: Kilduff & Brass (2010)

Small worlds

THEORETICAL PERSPECTIVES





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Small worlds: example

Theoretical Perspectives

Inter-firm alliance networks in 11 industries

Dense local clustering
fosters communication and cooperation and
therefore provides information transmission capacity in the network

Non-redundant connections contract the distance between firms
and therefore gives the network greater reach
tapping into a wider range of knowledge sources

Firms embedded in alliance networks with these properties exhibit greater patenting outcomes.

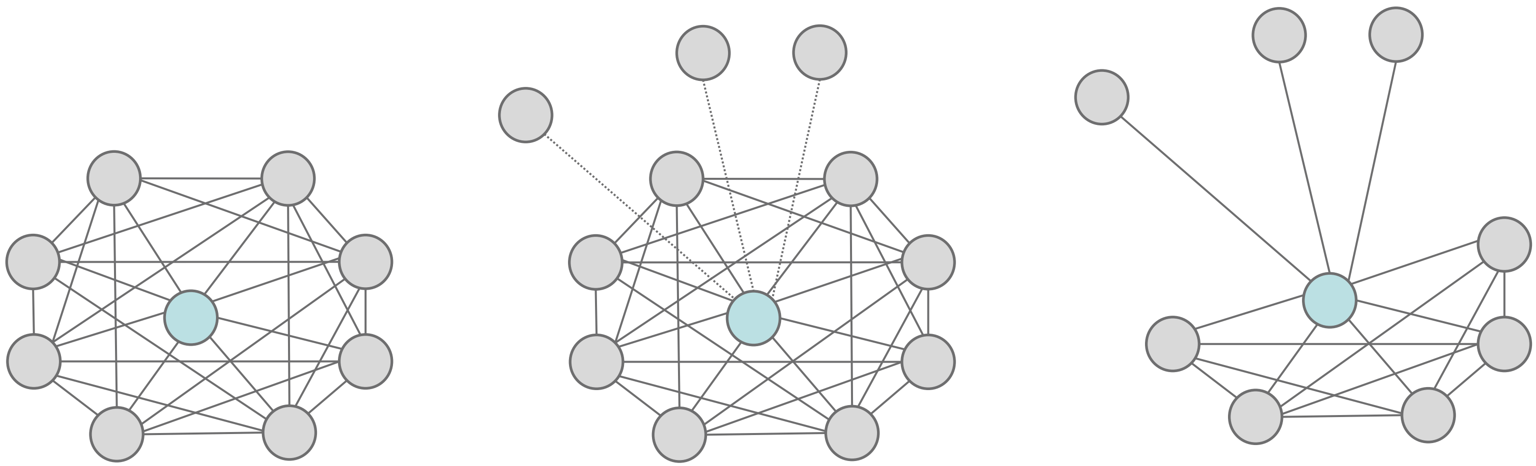
Source: Schilling & Phelps (2007, Man Sci)

Learning objectives

- …distinguish a range of existing concepts and arguments that explain how **networks** may **impact** on **innovation** ?✓
- …position these in the **major theoretical perspectives** and **research traditions** of the field of networks research ?✓
- …describe examples of the **current state-of-the art in the field**
- … with the ultimate aim of **positioning your research** against existing theories and state-of-the-art in the field

Strength of weak ties

Recent Advances in the Field



Firms take strategic actions to search for potential ties or reactivate latent (weak) ties in order to solve problems of network redundancy.

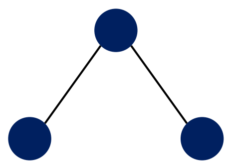
Source: Mariotti & Delbridge (2012, Org Sci) – Example for illustrative purposes only

Brokerage

RECENT ADVANCES IN THE FIELD

Tertius gaudens

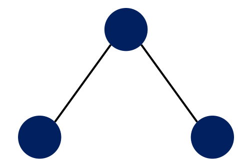
THE THIRD-WHO-BENEFITS



Behavioural orientation towards playing off people against each other for own benefit.

Tertius iungens

THE THIRD-WHO-JOINS



Behavioural orientation towards connecting disconnected individuals.

“Nexus work”

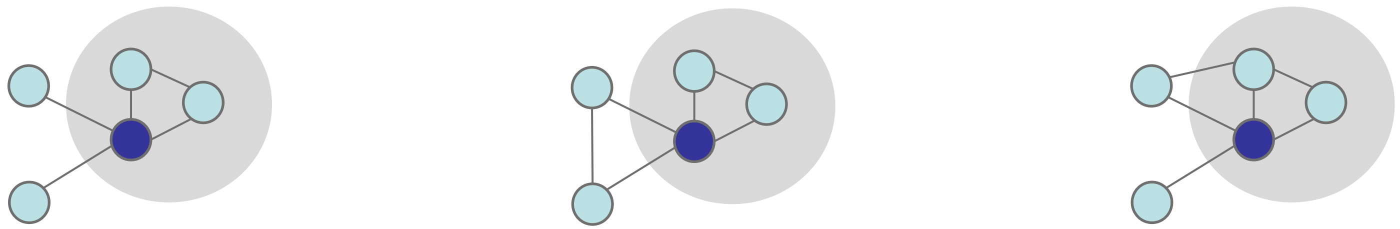
Tertius iungens orientation of engineers in an automotive predicts their involvement of individuals in innovative outcomes. The tertius iungens is a network process more than a network structure.

Source: Obstfeld (2005), Lingo & O’Mahony (2010)

Brokerage vs. boundary-spanning

Theoretical Perspectives

Brokers can span boundaries but not all boundary-spanners broker.



Organizational boundaries - Disciplinary boundaries – Sectoral boundaries

Brokers may suffer from a lack of trust or
may be sanctioned for self-interested behaviour.

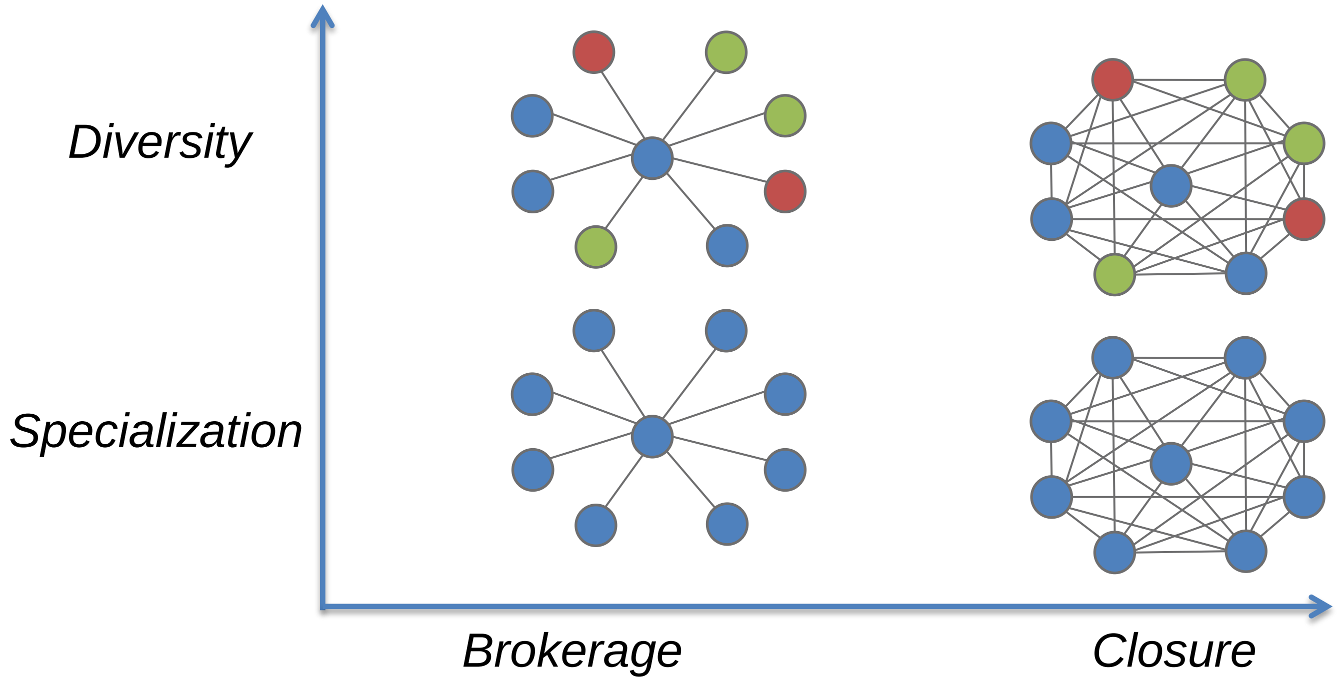
Boundary-spanners do not have such handicaps.

Source: Fleming et al. (2007), Fleming & Waguespack (2007)

Network range / actor diversity

RECENT ADVANCES IN THE FIELD

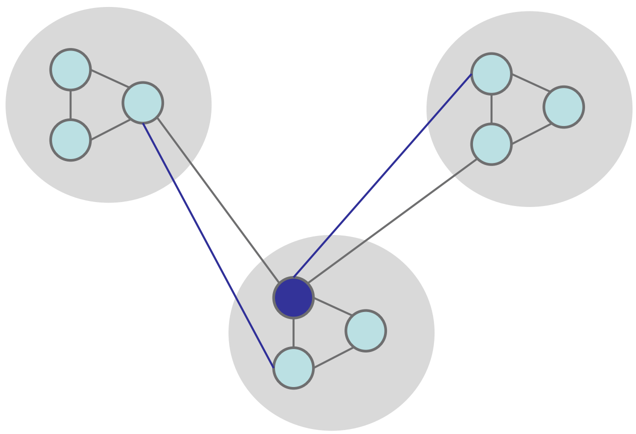
The brokerage-closure debate **implicitly assumes** that structural holes reflect content diversity.



Source: Ter Wal et al. (2013, WP) Reagans & McEvily (2003), Fleming et al. (2007), Zaheer & Soda (2009)

Simmelian ties

Recent Advances in the Field



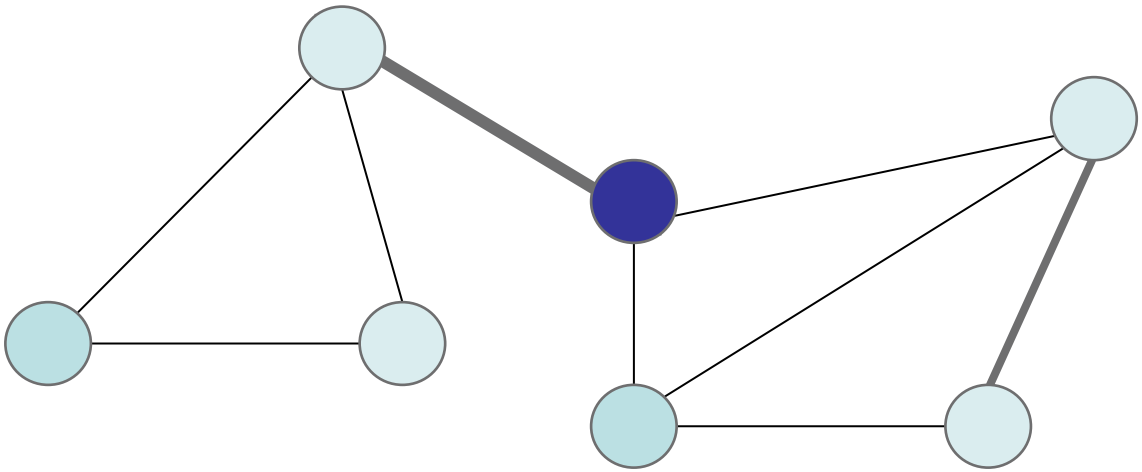
Ties that span organizational boundaries have no advantages for the generation of innovation per se.

Only when bridging ties are **Simmelian** do they promote innovation.

Source: Tortoriello et al. (2010)

Social capital

RECENT ADVANCES IN THE FIELD



Positive externalities of social capital describe whether an actor's social capital spills over to and improve the outcomes of those to whom he/she is connected.

Source: Galunic et al. (2012)

Learning objectives

- …distinguish a range of existing concepts and arguments that explain how **networks** may **impact** on **innovation** ✓
- …position these in the **major theoretical perspectives** and **research traditions** of the field of networks research ✓
- …describe examples of the **current state-of-the art in the field** ?✓
- … with the ultimate aim of **positioning your research** against existing theories and state-of-the-art in the field

Implications for your research

Some **advice** for positioning your own research:

- Choose **one** relevant theoretical stream to connect to.
  - Where does the phenomenon you study fit in the broader patchwork of theories?
  - What is the current state of understanding in that field?
  - What is the research gap?
  - Why is it worth filling?
- Demonstrate how your study **advances** or **challenges** existing theories.
  - Frame your theoretical reasoning exclusively in the theory of your choice.
  - Implications for theory section is a crucial element of a good paper.
  - Your study is context-specific, but what does it mean for the broader theoretical debate?

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